

Experiment No. 2

Title: Design and implement web application in a short sprint using scrum framework.

Problem Description: Students are expected to develop a 2-to-3-page web/mobile application by forming the team of 6 members.

The roles are to be decide by team members:

○ **Product owner-** course instructor/ 1 student from team. ○

Scrum master- manages the process, conduct meetings ○

Development team- Build and deliver product.

The Scrum process has to be managed using the agile tool – jira /Trello. Free versions.

Aim:

To develop a 2–3 page web/mobile application by applying Scrum framework and managing the sprint using an agile tool.

Tools Required:

- ✦ Agile Tool: Jira (Free version)
- ✦ Programming Tools: VS Code / Android Studio / Any IDE
- ✦ GitHub (optional for version control)
- ✦ Browser

Observation / Procedure

Step 1: Team Formation

1. Form a team of 6 members.
2. Assign roles:
 - ✦ Product Owner
 - ✦ Scrum Master
 - ✦ Development Team (4 members)

Step 2: Project Selection

1. Select a simple 2–3 page application. Examples: (Your project details)
 - ✦ Student Attendance System
 - ✦ To-Do List Application
 - ✦ Online Quiz System ✦ Mini E-Commerce Page
2. Clearly define:
 - ✦ Problem statement
 - ✦ Objective of application
 - ✦ Target users

Step 3: Create Agile Workspace

1. Create an account in Jira.
 2. Create a new project board.
 3. Choose Scrum template.
- Steps:
1. Set up the space. Name the space and then click get started

Welcome, anusha.ise! Set up your space

This is where you'll help your team track progress, stay organized, and manage tasks.

Name your space:

Cambridge Team

Example space names:

My Team

Team Luna

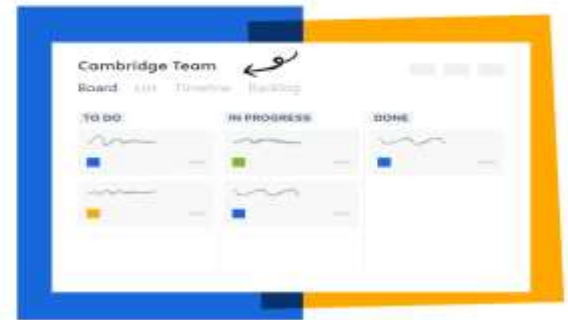
Task Tracking

Budget Planning

Big Idea Launch

Get started

Jira



2. Select the tasks that need to be done and click next.

What types of work do you need?

These form the building blocks of your space.

Feature
A broad piece of functionality.

Task
A small piece of work.

Story
A requirement expressed from the user's perspective.

Bug
A problem that needs fixing.

Request
An ask for assistance.

Don't worry, you can change these later.

Next

Jira



3. Track the work and click next.

How do you track work?

As you complete work, it moves through these statuses.

To Do

In Progress

In Review

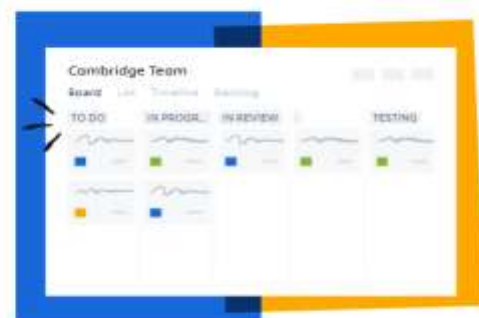
Name of status

Testing

Don't worry, you can change these later.

Next

Jira



4. Click on go to Jira and create a project.



Jira's better when used together

Jira works better with your team onboard. Invite a teammate to try it out with you.

Share via link

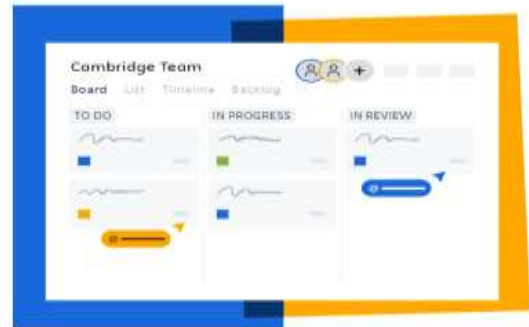
<https://cambridge-team-joi92b5z.atlassian.net?>

Copy link

Invite via email

e.g. maria@company.com

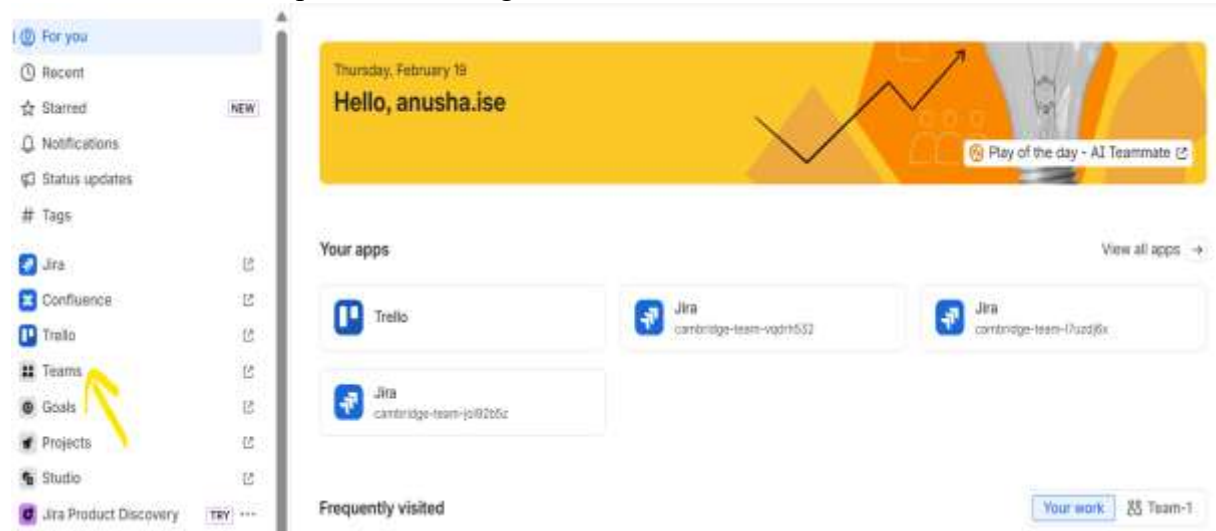
This site is protected by reCAPTCHA and the Google [Privacy Policy](#) and [Terms of Service](#) apply.



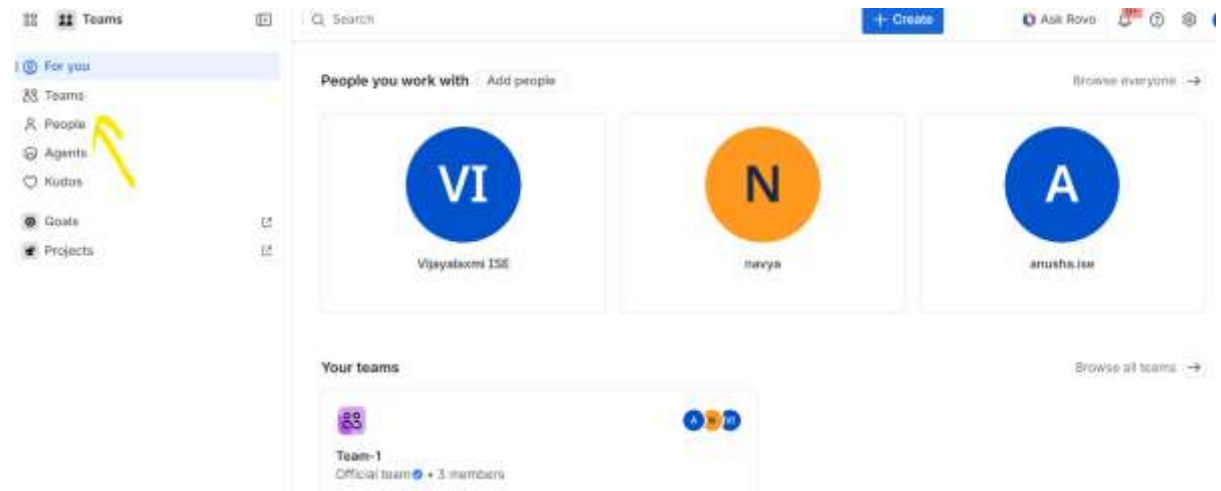
Do this later

Go to Jira

5. Select teams from options for adding the members into the team.



6. Add members to the team



7. Add members to the team by specifying the mail id of the members and click add.

Add people to Jira

Names or emails *

e.g., Maria, maria@company.com

This site is protected by reCAPTCHA and the Google [Privacy Policy](#) and [Terms of Service](#) apply.

Cancel Add

8. Go to Teams and create a team by giving name to the team and click create

4. Create columns:
 - ✚ To Do
 - ✚ In Progress
 - ✚ Testing
 - ✚ Done

Step 4: Create Product Backlog

1. Product Owner lists features as **User Stories**. Format:

As a user, I want to _____ so that _____.

Example:

- ✚ As a user, I want to login so that I can access my dashboard.
- ✚ As a user, I want to add tasks so that I can manage my work.

2. Add minimum 8–10 user stories.

Step 5: Sprint Planning

1. Decide sprint duration (3–5 days).
2. Select user stories for Sprint 1.
3. Break each story into tasks.
4. Assign tasks to team members.
5. Estimate effort (story points or simple priority).

Step 6: Conduct Daily Scrum

1. Conduct 10-minute daily meeting.
2. Each member answers:
 - ✦ What did I complete?
 - ✦ What will I do next? ✦ Any obstacles?
3. Scrum Master updates board status.

Step 7: Development Phase 1.

- Developers implement features.
2. Move task cards:
 - ✦ To Do → In Progress → Testing → Done
 3. Integrate modules.
 4. Test functionality.

Step 8: Sprint Review

1. Demonstrate completed product to Product Owner.
2. Verify if user stories are satisfied.
3. Record feedback.
4. Update backlog if needed.

Step 9: Sprint Reviewing Team

discusses:

- ✦ What went well?
- ✦ What problems occurred?
- ✦ What improvements are needed?

Scrum Master documents observations.

Step 10: Documentation

Students must include in record:

1. Mini project title and Abstract of your mobile/web Application
2. Team Members with Roles
3. Backlogs of your mobile/web Application.
4. Roles and responsibilities- Work done: to do list - work assigned /Task – Timeline – Testing - in progress - Completed (In Table Format)

5. Sprint Backlog
6. Board screenshots (Before and After Sprint)
7. Screenshots of developed application
8. Stand-up meeting points of your task/project/experiment progress - A small write up.
9. Mini project's coding and other required screen shots, etc - should be stored as data repository in GitHub.

Result:

The web/mobile application was successfully developed using Scrum framework within a short sprint and managed using an agile tool.